Ansys Fluent Simulation Report

Analyst	Admin
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Table of Contents

System Information

Application	Fluent
Settings	3d, double precision, pressure-based, SST k-omega
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:40:34 EDT
CPU	AMD Ryzen AI 9 HX 370 w/ Radeon
OS	Windows

Geometry and Mesh

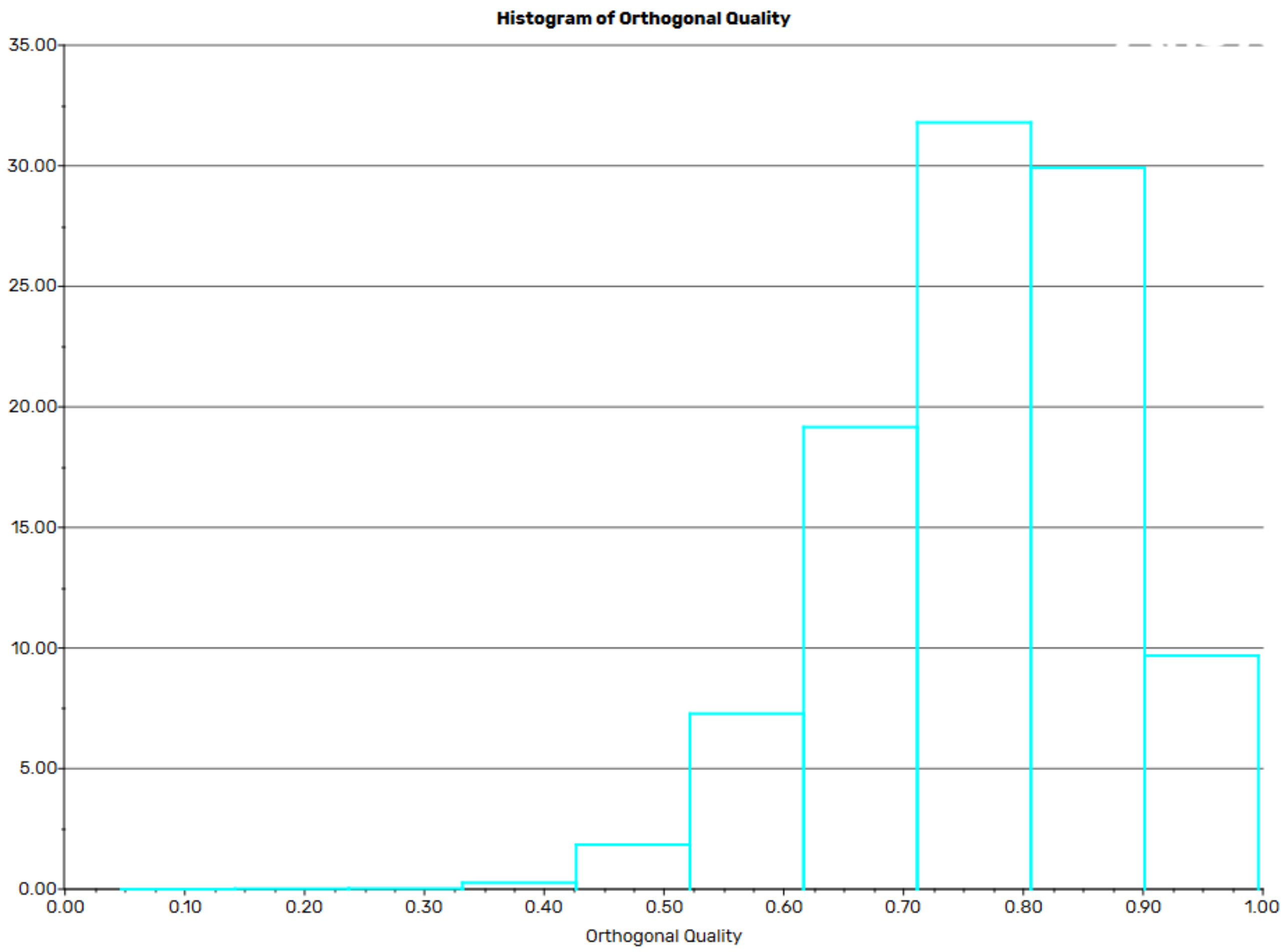
Mesh Size

Cells	Faces	Nodes
659527	1339391	115493

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
rotating	Tet Cell	0.047004127	47.326621
static	Tet Cell	0.34695183	10.639098

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	SST k-omega turbulence model

Material Properties

− Fluid	
− air	
Density	0.095 kg/m^3
Viscosity	1.34e-05 kg/(m s)
− Solid	
− carbonfibre-woven	
Density	1580 kg/m^3

Cell Zone Conditions

− Fluid	
− rotating	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	yes
Relative To Cell Zone	-1
Reference Frame Rotation Speed [rev/min]	-120
Reference Frame X-Velocity Of Zone [m/s]	0

Reference Frame Y-Velocity Of Zone [m/s]	0
Reference Frame Z-Velocity Of Zone [m/s]	0
Reference Frame Rotation-Axis Origin [m] (x,y,z)	(0, 0, 0)
Reference Frame Rotation-Axis Component (x,y,z)	(0, 0, 1)
Reference Frame User Defined Zone Motion Function	none
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no
– static	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

– Inlet	
– inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute
Velocity Magnitude [m/s]	0
Supersonic/Initial Gauge Pressure [Pa]	0
Turbulence Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	0.1
Turbulent Viscosity Ratio	1
– Outlet	
– outlet_	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0

Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	0.1
Backflow Turbulent Viscosity Ratio	1
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
− Wall	
− wall-14	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
− wall-13	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
− wall-static	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
− propeller	
Wall Motion	Stationary Wall

Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	rotating

Solver Settings

– Equations	
Flow	True
Turbulence	True
– Numerics	
Absolute Velocity Formulation	True
– Under-Relaxation Factors	
Pressure	0.3
Density	1
Body Forces	1
Momentum	0.7
Turbulent Kinetic Energy	0.8
Specific Dissipation Rate	0.8
Turbulent Viscosity	1

— Pressure-Velocity Coupling	
Type	SIMPLE
— Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	18.329 seconds
Iteration	237.769 seconds
AMG	63.216 seconds
Virtual Current Memory	4.70252 GB
Virtual Peak Memory	5.44261 GB
Memory Per M Cell	6.0991

Solution Status

Iterations: 506

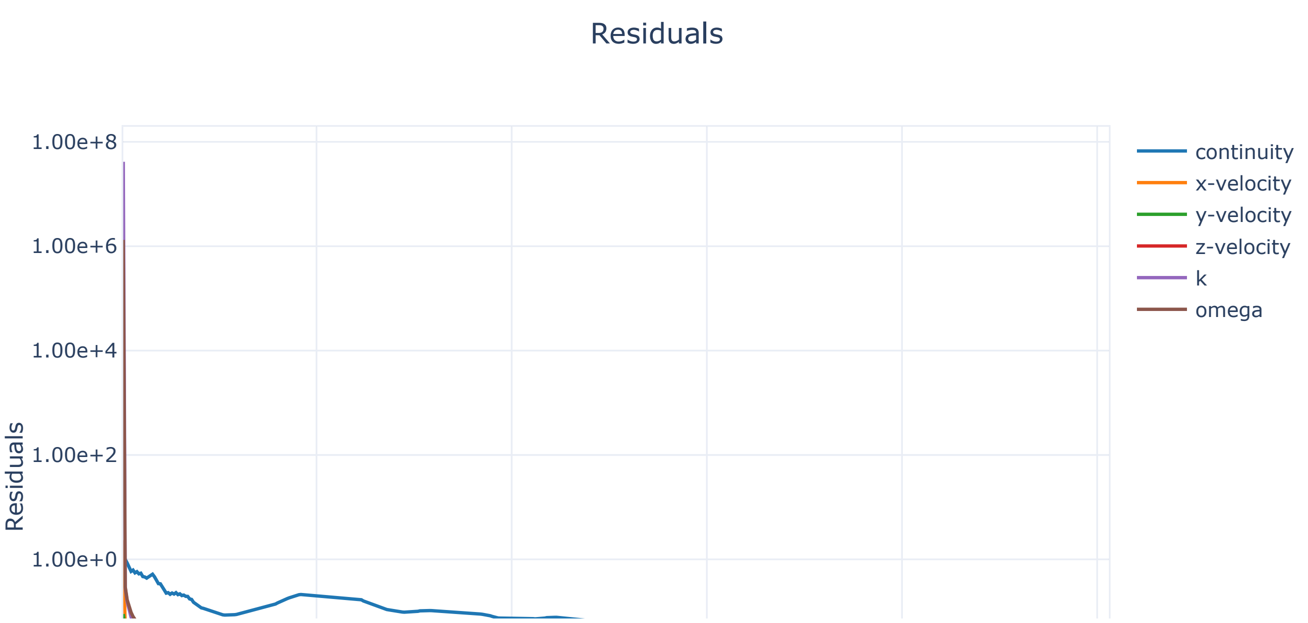
	Value	Absolute Criteria	Convergence Status
continuity	0.02201071	0.001	Not Converged
x-velocity	3.777368e-05	0.001	Converged
y-velocity	3.816965e-05	0.001	Converged
z-velocity	3.498087e-05	0.001	Converged
k	0.0001855581	0.001	Converged
omega	0.000123201	0.001	Converged

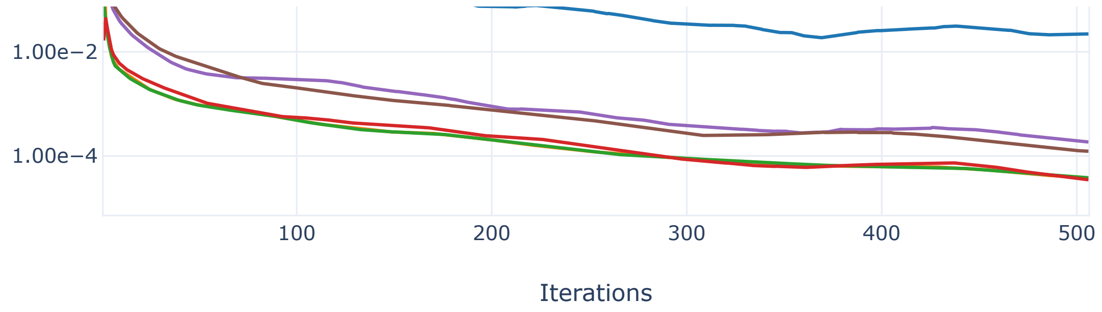
Report Definitions

report-def-lift	475.8845	N

Plots

Residuals

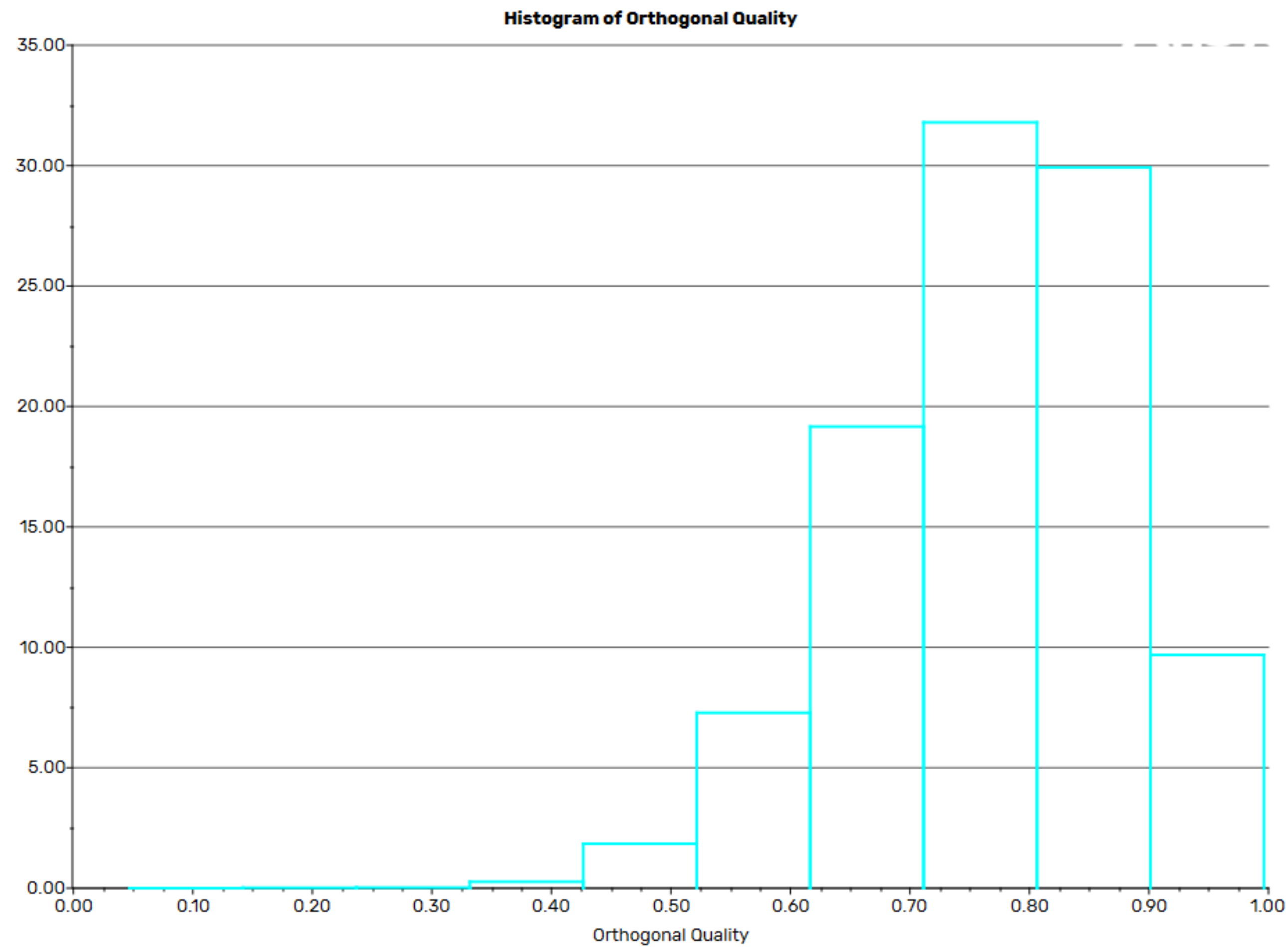


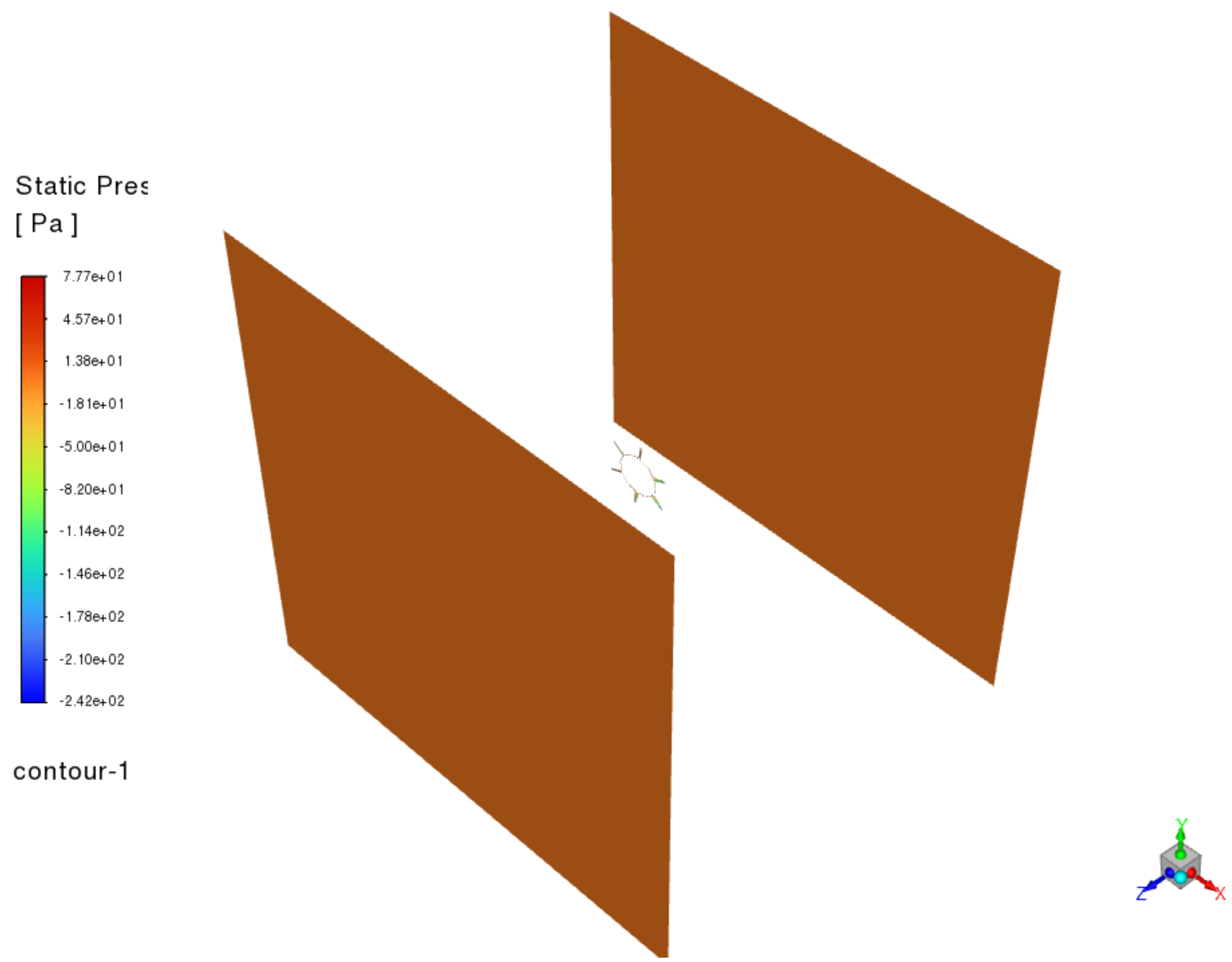


Contours

contour-1

Scenes


Histogram of Orthogonal Quality
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00
Orthogonal Quality


Static Pres
[Pa]
7.77e+01
4.57e+01
1.38e+01
-1.81e+01
-5.00e+01
-8.20e+01
-1.14e+02
-1.46e+02
-1.78e+02
-2.10e+02
-2.42e+02
contour-1
Y
Z
X